HYMAVATHI NALABOTHU
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PROFESSIONAL SUMMARY

- Designed and managed highly available infrastructure across **AWS**, **Azure**, and hybrid environments, ensuring fault tolerance and scalability for mission-critical applications while maintaining strict adherence to service level agreements (SLAs) and operational reliability standards.
- Implemented infrastructure as code using **Terraform**, **CloudFormation**, and **Ansible** to standardize deployments, reduce manual errors, and ensure repeatable, auditable, and maintainable infrastructure provisioning processes across multiple production environments.
- Monitored, troubleshot, and optimized distributed applications using **Prometheus**, **Grafana**, and **ELK Stack**, identifying performance bottlenecks and resolving latency issues to maintain system responsiveness and reliability under high-demand conditions.
- Developed and maintained CI/CD pipelines with **Git**, **Jenkins**, and container-based deployment frameworks, automating application releases and operational tasks to streamline delivery processes while reducing human intervention and deployment errors.
- Orchestrated and maintained **Kubernetes** clusters, **Docker** containers, and **Helm** deployments, ensuring seamless scaling, efficient resource utilization, and high availability for microservices and critical production workloads.
- Applied scripting expertise in **Python**, **Bash**, and **Go** to automate system maintenance, routine operational tasks, and monitoring integrations, improving efficiency and minimizing manual intervention across multiple infrastructure platforms.
- Managed cloud resources effectively in **AWS**, **Azure**, and **GCP**, ensuring secure configurations, cost optimization, and operational readiness for distributed systems while maintaining compliance with industry regulations such as HIPAA, PCI-DSS, and GDPR.
- Ensured security and compliance by implementing access controls, secret management, and auditing processes, protecting sensitive data and maintaining regulatory adherence across production and development environments using industry-standard tools and methodologies.
- Established and enforced SLOs, SLIs, and SLAs for critical systems, tracking performance metrics and implementing improvements to maintain reliability, uptime, and customer satisfaction across mission-critical applications.
- Collaborated with development, QA, and product teams to improve system reliability, scalability, and performance, sharing operational insights and guiding technical decisions that enhanced infrastructure efficiency and reduced incident recurrence.
- Mentored junior engineers and led technical initiatives to improve operational excellence, knowledge sharing, and adherence to best practices in monitoring, deployment, and infrastructure management.
- Administered databases including **PostgreSQL**, **MySQL**, **MongoDB**, and **Redis**, optimizing query performance, managing replication, and ensuring consistent availability for high-traffic applications and analytics workloads.
- Investigated and resolved incidents using **Splunk**, **ELK**, and **CloudWatch**, conducting post-mortem analyses, identifying root causes, and implementing preventive measures to reduce recurrence and improve overall system reliability.
- Integrated messaging and streaming platforms such as **Kafka**, **RabbitMQ**, and **Pub/Sub**, supporting event-driven architectures, ensuring high-throughput data processing, and maintaining consistent performance for real-time analytics and distributed workloads.

TECHNICAL SKILLS

Cloud Platforms: AWS, GCP, AZURE
Build Tools: Maven, ANT, Gradle
Operating Systems: Red Hat Linux, CentOS, Ubuntu, Windows, VMware and Mac OS.
Web/ Application Servers: Apache Tomcat, Nginx, JBoss, WebLogic, WildFly. **Databases**: MySQL, PostgreSQL, MongoDB, RDS, DynamoDB, Azure SQL Database **Version Control Tools**: Git, SVN, GitHub, Bitbucket
Iaac &Configuration Management: Terraform, Ansible, Chef, Puppet
CI/CD Tools: Jenkins, TeamCity, GitHub Actions, GitLab, XL Release, AWS Code deploy
Container Tools: Kubernetes, Docker, OpenShift, ARO
Monitoring & Logging Tools: Prometheus, Grafana, AWS CloudWatch, ELK Stack, Splunk
APM Tools: App Dynamics, Data Dog, New Relic

Scripting & Programming Languages: Java, Python, Ruby, Bash, PowerShell, JavaScript, HTML/CSS, JSP
Artifactory Management Tools: JFrog Artifactory, Nexus
Bug Tracking & Testing tools: JIRA, JUnit
Code Quality control tool: SonarQube, Checkmark
Other tools: Kafka, ServiceNow

CERTIFICATIONS

AWS Certified Solutions Architect – Associate
Microsoft Azure Administrator (AZ-104)

PROFESSIONAL EXPERIENCE

HCA Healthcare: November 2022 - Current (Location: Nashville, TN)

Site Reliability Engineer

- Designed and deployed **HIPAA-compliant** cloud infrastructure on **AWS** using **Terraform** and **AWS CloudFormation**, implementing **multi-region** deployments with **fault-tolerant architectures** for **EHR** systems and patient data processing across development, staging, and production environments with reusable **Terraform** modules.
- Established **SLOs**, **SLIs**, and error budgets for mission-critical healthcare services, maintained 99.95% uptime for patient-facing applications, developed **HIPAA-compliant** incident response protocols, and documented post-mortem analyses in **Confluence** to drive continuous operational improvement and reliability excellence.
- Built **CI/CD pipelines** using **Jenkins** and **Git** workflows, automated deployments for containerized healthcare microservices with **pytest**, implemented **blue-green deployment** strategies, and enforced security scanning and compliance validation using **Python** and **Bash** scripting before production releases.
- Orchestrated **Kubernetes** clusters on **AWS EKS** and **OpenShift**, deployed **Docker** containers using **Helm** charts, implemented **HIPAA-compliant** pod security policies and network segmentation, and maintained cluster health through automated monitoring and capacity planning with **Python** scripts integrated with **AWS CloudWatch**.
- Implemented comprehensive observability systems using **Prometheus**, **Grafana**, **AWS CloudWatch**, and **AppDynamics**, configured **Alertmanager** for proactive incident detection, deployed **Prometheus** exporters across **Linux** infrastructure, and established custom dashboards tracking **SLIs**, **SLOs**, latency, and error rates for real-time visibility.
- Designed and enforced **HIPAA** and **HITECH** security controls, implemented **AWS IAM** policies and **RBAC**, configured **AWS KMS** for secret management and encryption at rest and in transit using **SSL/TLS** certificates, and established audit logging to **CloudWatch Logs** for all patient data workflows.
- Automated operational tasks using **Python**, **Bash**, and **Go** scripts for backup validation, log rotation, certificate renewal, and infrastructure health checks, developed **Ansible** frameworks for configuration management, and standardized security baselines across **AWS** cloud infrastructure and on-premise **Linux** servers.
- Managed and optimized **AWS** services including **AWS EC2**, **AWS RDS**, **AWS Lambda**, **AWS S3**, and **AWS Route 53**, configured **multi-AZ** deployments with disaster recovery capabilities, deployed **PostgreSQL**, **MongoDB**, and **Redis** clusters with automated backups, replication, and **CloudWatch** monitoring.
- Implemented event-driven architectures using **AWS Lambda**, **AWS SQS/SNS**, **Kafka**, and **RabbitMQ** for processing healthcare event streams and patient notifications, designed **Kafka** streaming pipelines for real-time data ingestion, and integrated **Pub/Sub** messaging patterns to decouple microservices while maintaining **HIPAA compliance**.
- Configured load balancing and reverse proxy solutions using **Nginx** and **AWS Route 53** with health checks, implemented **SSL/TLS** termination, DNS failover, and geo-routing policies, managed **Nginx** API gateway configurations, and integrated **AWS CloudWatch** for access logging and performance monitoring supporting troubleshooting.
- Conducted in-depth troubleshooting and performance optimization of distributed systems, analyzed latency issues and network bottlenecks in **Linux** environments using **AppDynamics** and **Prometheus**, resolved critical **Kubernetes** pod failures, **PostgreSQL** database deadlocks, and networking issues applying deep **TCP/IP** protocol knowledge.
- Designed disaster recovery and backup strategies for **AWS S3**, **AWS RDS**, **MongoDB**, and **Redis**, ensured **RTO** and **RPO** aligned with **HIPAA** requirements, automated backup validation and restoration testing using **Python** scripts integrated with **Jenkins** to maintain data integrity and patient data security standards.
- Led incident management processes and facilitated post-mortem analysis sessions using **JIRA** and **Confluence**, documented root causes and corrective actions, established incident response playbooks and on-call rotations, and integrated **Alertmanager** with **PagerDuty** for escalation during critical production incidents affecting patient care.
- Enhanced monitoring and logging infrastructure by deploying centralized log aggregation using **CloudWatch Logs**, implemented retention policies and alerting rules, configured **AWS CloudWatch** dashboards and custom metrics for **AWS Lambda**, **AWS EKS**, and **AWS RDS** tracking performance, resource utilization, and cost optimization opportunities.
- Collaborated with development and QA teams to define release management best practices, conducted code reviews and infrastructure design sessions using **Git** workflows, mentored junior engineers on **SRE** principles, **Terraform** module development, and **Kubernetes** best practices to promote operational excellence across the organization.

Environment: Confluence, Aws Kms, Appdynamics, Aws Cloudwatch, Aws Eks, Aws Ec2, Nginx, AWS S3, Rabbitmq, Aws Route 53, Bash, Jenkins, AWS, Aws Rds, Linux, Openshift, Kafka, Cloudwatch, Mongodb, Pytest, Terraform, Ansible, Kubernetes, Python, Git, Aws Cloud Formation, Alertmanager, Postgresql, Redis, Grafana, Aws Lambda, Aws Iam, Prometheus, Docker, Go, Amazon SQS/SNS, Jira.

American Express: January 2020 - October 2022 (Location: New York, NY)

Site Reliability Engineer

- Designed and deployed **PCI-DSS**-compliant **cloud infrastructure** on **Azure** using **Terraform** and **ARM templates** for **IaC** provisioning, implementing **fault-tolerant** architectures with **Azure Kubernetes Service**, **Azure Virtual Machines**, and **HAProxy** load balancing to achieve **99.95% SLA** across multi-region banking transaction systems.
- Established **SRE principles** including **SLIs**, **SLOs**, and **error budgets** for banking applications, implementing **monitoring** frameworks with **Prometheus**, **Grafana**, **Azure Monitor**, and **Zabbix** to track reliability metrics, while defining **incident response** workflows that reduced **MTTR** and improved operational transparency.
- Built **CI/CD pipelines** using **GitHub Actions** and **Git** workflows for automated deployment of containerized microservices to **Kubernetes** and **OpenShift** clusters, incorporating **Helm** chart management, **Python** and **Bash** validation scripts, and automated **unittest** execution with comprehensive rollback capabilities.
- Orchestrated **distributed systems** for real-time payment processing using **Docker** containers on **Azure Kubernetes Service** with **Helm** deployments, configuring **event-driven** messaging through **Azure Service Bus**, **Kafka**, and **RabbitMQ** with circuit breaker patterns ensuring data integrity across microservices communication.
- Implemented enterprise **observability** using **ELK Stack**, **Splunk**, **CloudWatch**, **Prometheus**, and **Alertmanager** for centralized **logging** and metric collection, enabling proactive detection of **performance bottlenecks** and **latency issues** in banking transactions while maintaining **audit controls** aligned with regulatory requirements.
- Hardened **security** across **Linux** infrastructure by configuring **Azure Active Directory** role-based authentication, managing secrets through **Azure Key Vault** with automated rotation, deploying **Azure Firewall** and **Azure VPN** with segmented **subnets**, and enforcing **PCI-DSS** compliance including vulnerability scanning and privileged access audit logging.
- Automated operational tasks through **Python** and **Bash** scripting including backup orchestration to **Azure Blob Storage**, health checks for **Azure SQL Database**, **MySQL**, **MongoDB**, and **Redis** instances, certificate renewal workflows, and **Terraform** state validation to eliminate configuration drift across environments.
- Deployed **serverless architectures** using **Azure Functions** for event-driven fraud detection and compliance reporting, integrating **Azure Service Bus** and **Pub/Sub** messaging patterns to improve **scalability**, while optimizing function performance to maintain sub-second response times for customer-facing transaction validation services.
- Conducted **troubleshooting** and root cause analysis of **distributed systems** failures, resolving **networking** bottlenecks, database connection issues in **PostgreSQL** and **MySQL**, **Redis** cache invalidation, **Kubernetes** orchestration failures, and cross-region **Azure** latency using systematic diagnostic methodologies including distributed tracing.
- Led **incident management** processes including real-time coordination, stakeholder communication, and comprehensive **post-mortem analysis** documentation for **Sev1** and **Sev2** incidents, facilitating blameless retrospectives that identified systemic weaknesses and drove implementation of preventive controls and **monitoring** enhancements strengthening **site reliability**.
- Optimized **performance** of high-throughput banking applications through database query tuning in **MySQL** and **Azure SQL Database**, **Redis** caching refinement, **Kafka** consumer rebalancing, and **Azure Monitor** profiling with **Python** instrumentation, ensuring consistent sub-100ms response times for payment authorization workflows.
- Maintained hybrid **cloud** and on-premise environments integrating legacy banking systems with **Azure** infrastructure through **Azure VPN** gateways, configuring **networking** topologies with **Azure Firewall**, and synchronizing **MongoDB** clusters with **Azure Blob Storage** for disaster recovery aligned with **PCI-DSS** mandates.
- Implemented comprehensive **disaster recovery** strategies across **Azure** regions, designing automated failover for **Azure Kubernetes Service** workloads, **Azure SQL Database** geo-replication, and stateful recovery procedures, conducting quarterly drills validating **RTO/RPO** targets and testing **Azure Blob Storage** restoration processes per regulatory requirements.
- Developed custom **Prometheus** exporters in **Python** for banking-specific metrics including transaction rates and gateway latency, visualizing data through **Grafana** dashboards, configuring **Alertmanager** routing to **PagerDuty**, and establishing alert tuning practices minimizing false positives while ensuring rapid **incident response** for customer-facing operations.

Environment: Azure, Terraform, ARM Templates, Azure Kubernetes Service (AKS), Azure Virtual Machines, HAProxy, Prometheus, Grafana, Azure Monitor, Zabbix, GitHub Actions, Git, Kubernetes, OpenShift, Helm, Docker, Python, Bash, Azure Service Bus, Kafka, RabbitMQ, ELK Stack (Elasticsearch, Logstash, Kibana), Splunk, Amazon CloudWatch, Alertmanager, Azure Active Directory (AAD), Azure Key Vault, Azure Firewall, Azure VPN Gateway, Azure Blob Storage, Azure SQL Database, MySQL, PostgreSQL, MongoDB, Redis, Azure Functions, PagerDuty.

Best Buy: March 2017 - December 2019 (Location: Richfield, MN)

Cloud DevOps Engineer

- Designed and deployed scalable cloud infrastructure on **Google Cloud Platform (GCP)** using **Compute Engine**, **Google Kubernetes Engine (GKE)**, **Cloud SQL**, and **Cloud Storage** for retail e-commerce applications. Implemented **infrastructure as code** with **Terraform** to automate provisioning across environments while maintaining **GDPR** and **PCI-DSS** compliance for transaction security.
- Established comprehensive **SLIs**, **SLOs**, and error budgets for retail services ensuring 99.95% uptime, monitored using **Prometheus** for metrics collection, **Grafana** for visualization, and **Cloud Monitoring** for resource tracking. Enabled proactive identification and resolution of performance bottlenecks and latency issues affecting customer checkout workflows during peak periods.
- Built **CI/CD pipelines** using **GitLab CI/CD** integrated with **Git** workflows to automate deployments for containerized microservices across **Kubernetes** clusters, reducing deployment time significantly. Developed automation scripts in **Python** and **Bash** for infrastructure provisioning, **Terraform** state management, **Docker** image builds, and **Helm** chart deployments.
- Orchestrated **Google Kubernetes Engine (GKE)** clusters running **Docker** containers for 40+ microservices supporting retail operations including product catalog and shopping cart systems. Implemented **Helm** charts for standardized deployments, configured **Envoy** service mesh for traffic management, and established pod autoscaling policies based on CPU and memory metrics.
- Designed comprehensive observability systems using **Prometheus** for time-series metrics, **Grafana** dashboards for real-time visualization, **Cloud Logging** for centralized aggregation, and **Loki** for log querying. Configured intelligent alert routing through **Slack** and **MS Teams** with escalation policies tied to **SLO** thresholds, reducing mean time to resolution.
- Managed **GCP** resources including **VPC** network design, **Cloud DNS** configuration, **Google Cloud Storage** buckets, **Cloud SQL** for **PostgreSQL** and **MySQL** databases, and **Cloud Functions** for serverless processing. Optimized cloud infrastructure costs through rightsizing recommendations and lifecycle policies while maintaining high availability across critical retail services and applications.
- Implemented robust security controls using **Cloud IAM** for role-based access control, **GCP Secret Manager** for centralized secret management of API keys and database credentials, ensuring **GDPR** and **PCI-DSS** compliance. Configured access controls with least privilege principles, established audit logging, and implemented network security policies including firewall rules and VPN tunnels.
- Built event-driven systems leveraging **Google Pub/Sub** for asynchronous messaging supporting real-time inventory updates and order processing workflows across the retail platform. Integrated **Kafka** for high-throughput event streaming between legacy on-premise systems and cloud-native microservices, ensuring data consistency across distributed systems and financial transactions.
- Performed deep troubleshooting and performance optimization of distributed applications running on **Linux** systems, analyzing networking bottlenecks and database query patterns in **PostgreSQL**, **MongoDB**, and **Redis**. Resolved critical production incidents including memory leaks and connection pool exhaustion through systematic incident response procedures and detailed root cause analysis.
- Configured and optimized **PostgreSQL** and **MySQL** databases on **Cloud SQL** with read replicas and automated backups, implemented **MongoDB** clusters for product catalog storage, and **Redis** for session management. Enhanced database performance through query optimization, index tuning, and connection pooling, ensuring sub-second response times for critical retail operations.
- Developed serverless architectures using **Cloud Functions** triggered by **Pub/Sub** events and **Google Cloud Storage** changes to process order confirmations and send customer notifications without managing infrastructure. Implemented event-driven patterns for decoupled system communication, reducing operational complexity and enabling independent scaling while maintaining cost efficiency through pay-per-execution pricing.
- Established comprehensive **CI/CD** best practices including **Git** branching strategies and deployment strategies such as blue-green deployments within **GitLab CI/CD** pipelines for retail applications. Integrated **Selenium** for automated end-to-end testing of critical user journeys including checkout flows, ensuring code quality before production releases while maintaining deployment frequency targets.

Environment: Terraform, Python, Linux, Bash, puppet, Kubernetes, Selenium, PostgreSQL, Docker, Prometheus, mesos, GCP, Redis, Git, Grafana, Google Kubernetes Engine, MongoDB, GitLab CI/CD, Kafka, Envoy, Cloud SQL, GCP Secret Manager, Google Pub/Sub, loki, Google Cloud Storage, Cloud Functions, Compute Engine, Pub/Sub.

Progressive: December 2015 - February 2017 (Location: Mayfield Village, OH)

DevOps Engineer

- Designed and managed cloud solutions across AWS and Azure platforms, successfully deploying secure and scalable applications using critical services including AWS EC2, S3, RDS, Azure Virtual Machines, Blob Storage, and Azure SQL Database.
- Enhanced Software Development Life Cycle (SDLC) by implementing integration with Jenkins, GitHub Actions, and Azure DevOps for CI/CD pipelines, which automated build and deployment processes and reduced release cycles by 25%.
- Managed software artifact repositories using JFrog/Nexus Artifactory to ensure efficient storage, version control, and distribution across multiple environments.
- Streamlined administrative tasks with Groovy scripts in AWS, reducing manual efforts by 20% and generating Agile metrics in JIRA to drive data-driven decisions for team performance and productivity optimization.
- Administered MySQL and MongoDB databases on AWS RDS and Azure Cosmos DB, maintaining high availability, performing critical performance tuning, and implementing automated backup strategies for scalable and secure data management.
- Implemented robust access management by configuring Identity and Access Management (IAM) roles and policies in AWS and

Azure, deploying SSL certificates for encryption, and enforcing Role-Based Access Control (RBAC) to ensure fine-grained security controls.

- Utilized AWS Key Management Service (KMS) and Azure Key Vault for comprehensive key management, developing disaster recovery strategies and network security configurations to guarantee system security and regulatory compliance.
- Created and automated infrastructure management scripts using Shell and Python to efficiently handle resources across AWS and Azure environments, reducing manual intervention by 40% and enabling seamless cloud operations.
- Implemented comprehensive multi-cloud observability and performance monitoring through Datadog and New Relic, developing custom dashboards to track critical AWS metrics including CPU utilization, memory consumption, and latency, while using AWS CloudWatch for real-time monitoring and alerting to ensure proactive issue resolution and optimal system performance.

Environment: Aws, Azure, CI/CD, Git, Jfrog Artifact, JIRA, Ansible playbook, Jenkins pipeline, DevOps methodologies, MySQL, MongoDB, AWS RDS, AWS CloudWatch, IAM, New Relic, Datadog, Shell scripting, Python.

Mastek: June 2014 - October 2015 (Location: Mumbai, India)

Linux Administrator / System Engineer

- Administered and automated Linux/Unix environments, specifically Red Hat Enterprise Linux (RHEL), using Bash and Python scripts to enhance operational efficiency by 25%, streamlining system management and reducing manual intervention.
- Managed complex system configurations, including Logical Volume Management (LVM) setups, to optimize storage and backup processes, successfully reducing system downtime by 30% through strategic volume management and efficient data organization.
- Configured and maintained critical network services including DHCP, DNS, and FTP, while implementing robust security protocols such as SSL/TLS and IPSec to ensure secure and reliable network operations across infrastructure.
- Implemented high availability solutions using Red Hat Clusters and HAProxy, significantly improving system uptime and minimizing service interruptions by 40%, ensuring continuous operational reliability.
- Developed automated backup strategies and scheduled cron jobs for comprehensive disaster recovery, maintaining data integrity and enabling quick system recovery during potential failures. Implemented extensive system hardening techniques that strengthened security infrastructure and reduced system vulnerabilities by 40%.
- Performed advanced kernel parameter tuning to optimize system performance, increasing reliability and responsiveness in high-demand computing environments.
- Managed user and group permissions with SSH key authentication, implementing fine-grained access controls to secure server environments and protect critical system resources.
- Executed regular system maintenance by patching and updating Linux servers using RPM, YUM, and APT package management tools, maintaining system stability and proactively reducing potential security vulnerabilities.
- Configured Apache and Nginx web servers to ensure scalable, secure, and efficient service delivery for mission-critical applications, optimizing web infrastructure performance.
- Integrated Oracle databases with Linux environments, providing comprehensive database administration and performance optimization. Deployed and managed virtual environments using VMware and KVM, improving resource utilization and scalability across multiple data centers.

Environment: Red Hat Enterprise Linux, Bash, Python, Logical Volume Manager, Red Hat Cluster Servers, Oracle, MySQL, DNS, DHCP, FTP.

EDUCATION

- Bachelor of Technology Computer Science | **Year of passed:**